Automated Hand-Drawn Logic Circuit to Truth Table Converter Using Digital Image Processing

A PROPOSAL PRESENTED BY

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INTRODUCTION

Designing and testing digital logic circuits is a fundamental aspect of computer science and electrical engineering. Traditionally, students and engineers conceptualize these designs by hand-drawing logic gates and wires on paper. However, verifying the logic of these physical sketches requires either manually calculating the truth table—a tedious and error-prone process for complex circuits—or rebuilding the entire circuit from scratch within a digital simulation software suite.

The problem lies in the disconnect between the rapid, intuitive nature of physical sketching and the rigorous mathematical verification required for logic design. Existing Optical Character Recognition (OCR) and object detection models often rely heavily on heavy machine learning frameworks and struggle with the high variance of hand-drawn sketches, such as uneven lighting, paper textures, and inconsistent wire thicknesses. There is a need for a lightweight, deterministic image processing pipeline that can automatically digitize and simulate hand-drawn logic circuits without relying on black-box AI models.

OBJECTIVES

- To develop a robust preprocessing pipeline capable of isolating faint, hand-drawn ink from physical paper backgrounds under varying lighting conditions.
- To accurately segment and classify different types of hand-drawn logic gates (e.g., AND, OR) using geometric feature extraction (e.g., Convex Hull Solidity) rather than machine learning.
- To extract the circuit's topological netlist by mathematically skeletonizing the drawn wires and isolating connection endpoints.
- To map the extracted netlist into a Directed Acyclic Graph (DAG) and simulate the electrical flow to automatically generate a complete and accurate Truth Table.

APPROACH

The project will rely purely on fundamental Digital Image Processing (DIP) techniques, utilizing Python and OpenCV. The methodology is divided into five sequential phases:

1. **Preprocessing & Binarization:** The image will be converted to grayscale and processed using Adaptive Gaussian Thresholding to eliminate uneven shadows and isolate the ink. Morphological closing operations will be used to heal broken pen strokes.

2. **Gate Solidification & Isolation:** To separate gates from connecting wires, a "Flood Fill" algorithm will be applied to the closed loops of the gate shapes, converting hollow drawings into solid blobs. Morphological opening with calibrated kernels will then be used to erase the thin wires, leaving only the solid gate bodies.
3. **Feature-Based Classification:** Isolated gates will be classified using geometric properties. By calculating the ratio of the gate's contour area to its Convex Hull area (Solidity), the system will distinguish between shapes mathematically (e.g., a blocky AND gate yields a high solidity >0.90, while a curved OR gate yields a lower solidity <0.90).
4. **Skeletonization & Netlist Generation:** The isolated wires will be processed using the Zhang-Suen Thinning Algorithm to reduce them to a precise 1-pixel width. A custom hit-or-miss kernel will count pixel neighbors to identify exact endpoints. Connected Components labeling will then be used to map which wire connects to which gate input/output.
5. **Graph Traversal Simulation:** The visual topology will be converted into a Directed Acyclic Graph (DAG) data structure. The system will iterate through all possible binary input combinations, applying native bitwise operators at each gate node to calculate and output the final Truth Table.

Data Collection Strategy: To ensure the pipeline is robust against real-world variance, a custom dataset of hand-drawn circuits will be manually generated. I will draw a minimum of 30 distinct logic circuits (ranging from simple 2-gate setups to complex 10-gate topologies). These will be drawn using various writing instruments (blue ballpoint, black ink, pencil) on different substrates (blank paper, ruled notebooks). The images will be captured using a standard smartphone camera under varying real-world lighting conditions (daylight, harsh desk lamps, shadows) to thoroughly stress-test the adaptive thresholding and segmentation algorithms.

REFERENCES

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